Digital Logic Assignment Solutions

Assignment 3

December 24, 2025

1 Sequential Circuit Analysis with JK Flip-Flops

1.1 Derive the state equations

The characteristic equation for a JK flip-flop is $Q(t+1) = JQ' + K'Q$. Given the input equations:

$$\text{Flip-flop A:} \quad J_A = x, \quad K_A = B$$
$$\text{Flip-flop B:} \quad J_B = x, \quad K_B = A'$$

Substitute these into the characteristic equations to obtain the state equations for $A(t+1)$ and $B(t+1)$:

$$\begin{aligned} A(t+1) &= J_A A' + K_A' A \\ &= xA' + B'A \end{aligned}$$

$$\begin{aligned} B(t+1) &= J_B B' + K_B' B \\ &= xB' + (A')'B \\ &= xB' + AB \end{aligned}$$

(Another solution is to calculate FlipFlop input from input equation, then induce the next state. Please refer to lec7 page 16)

1.2 Draw the state table and state diagram

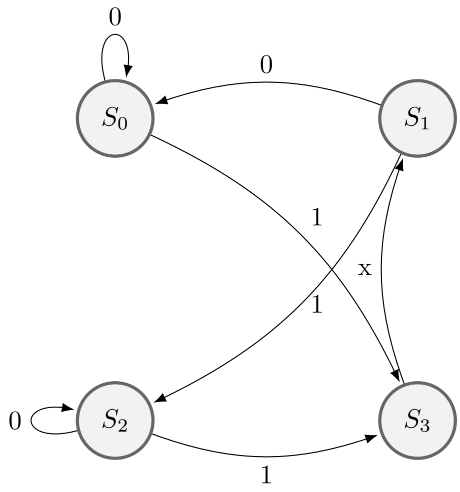
State Table:

The state table includes the intermediate values for the flip-flop inputs based on the equations derived: $J_A = x, K_A = B, J_B = x, K_B = A'$.

A	B	x	JA	KA	JB	KB	A(t+1)	B(t+1)
0	0	0	0	0	0	1	0	0
0	0	1	1	0	1	1	1	1
0	1	0	0	1	0	1	0	0
0	1	1	1	1	1	1	1	0
1	0	0	0	0	0	0	1	0
1	0	1	1	0	1	0	1	1
1	1	0	0	1	0	0	0	1
1	1	1	1	1	1	0	0	1

State Diagram:

States are encoded as: $S_0 = 00$, $S_1 = 01$, $S_2 = 10$, $S_3 = 11$.



2 Sequential Circuit Analysis (Problem 2)

2.1 Derive the functions

Based on the provided circuit diagram:

a) Input Functions

Tracing the connections to the T inputs of flip-flops A and B:

$$a1) \quad T_A = A + B$$
$$a2) \quad T_B = A' + B$$

b) State Functions (Next State Equations)

The characteristic equation for a T flip-flop is $Q(t+1) = T \oplus Q$. Substitute the input functions derived above:

For Flip-Flop A:

$$\begin{aligned} b1) \quad A(t+1) &= T_A \oplus A \\ &= (A+B) \oplus A \\ \text{Using } x \oplus y &= x'y + xy' : \\ &= (A+B)'A + (A+B)A' \\ &= (A'B')A + (AA' + BA') \\ &= 0 + 0 + A'B \\ &= \mathbf{A'B} \end{aligned}$$

For Flip-Flop B:

$$\begin{aligned} b2) \quad B(t+1) &= T_B \oplus B \\ &= (A'+B) \oplus B \\ &= (A'+B)'B + (A'+B)B' \\ &= (AB')B + (A'B' + BB') \\ &= 0 + A'B' + 0 \\ &= \mathbf{A'B'} \end{aligned}$$

c) Output Functions

The outputs are non-existent

d) State Table

The table lists the present state, the derived flip-flop inputs (T_A, T_B), and the next state calculated using the state functions. Since there is no external input x, that column is omitted.

A	B	TA	TB	A(t+1)	B(t+1)
0	0	0	1	0	1
0	1	1	1	1	0
1	0	1	0	0	0
1	1	1	1	0	0

3 Problem 3

3.1 Derive Equations

Based on the circuit diagram correction:

- Top Flip-Flop: Q_1
- Bottom Flip-Flop: Q_2
- Output Logic: XOR gate inputs are connected to X and Q_2'.

a1-a4) Input Functions

$$\begin{aligned} a1) \quad & J_1 = X \\ a2) \quad & K_1 = (XQ_2')' = X' + Q_2 \\ a3) \quad & J_2 = X \\ a4) \quad & K_2 = (XQ_1)' = X' + Q_1' \end{aligned}$$

a5-a6) State Equations

The characteristic equation for a JK flip-flop is $Q(t+1) = JQ' + K'Q$.

For $Q_1(t+1)$:

$$\begin{aligned} Q_1(t+1) &= XQ_1' + (X' + Q_2)'Q_1 \\ &= XQ_1' + XQ_2'Q_1 \\ &= X(Q_1' + Q_2') \end{aligned}$$

For $Q_2(t+1)$:

$$\begin{aligned} Q_2(t+1) &= XQ_2' + (X' + Q_1')'Q_2 \\ &= XQ_2' + XQ_1Q_2 \\ &= X(Q_2' + Q_1) \end{aligned}$$

a7) Output Function

The XOR gate inputs are X and Q_2'.

$$F = X \oplus Q_2'$$

3.2 State Table and Diagram

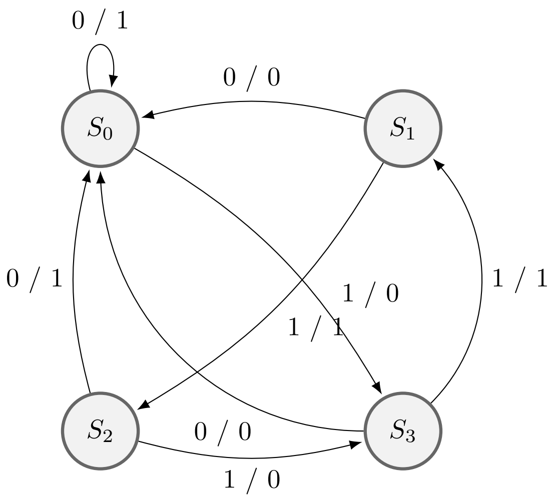
a8) State Table (Sorted by X, Q1, Q2)

Input/Present State			FF Inputs				Next State		Out
X	Q1	Q2	J1	K1	J2	K2	Q1(t+1)	Q2(t+1)	F
0	0	0	0	1	0	1	0	0	**1**
0	0	1	0	1	0	1	0	0	**0**
0	1	0	0	1	0	1	0	0	**1**
0	1	1	0	1	0	1	0	0	**0**
1	0	0	1	0	1	1	1	1	**0**
1	0	1	1	1	1	1	1	0	**1**
1	1	0	1	0	1	0	1	1	**0**
1	1	1	1	1	1	0	0	1	**1**

a9) State Diagram (Mealy Machine)

Since the output F depends on input X, we use a Mealy-style diagram where outputs are written on the transitions: **Input X / Output F**.

States: $S_0 = 00$, $S_1 = 01$, $S_2 = 10$, $S_3 = 11$.



b) Machine Type

This is a **Mealy Machine**. The output function $F = X \oplus Q_2'$ depends directly on the external input variable X as well as the state variable Q_2.

3.3 c) Timing Diagram

Input Sequence: $X = 101100$. Initial State: $Q_1 = Q_2 = 0$.

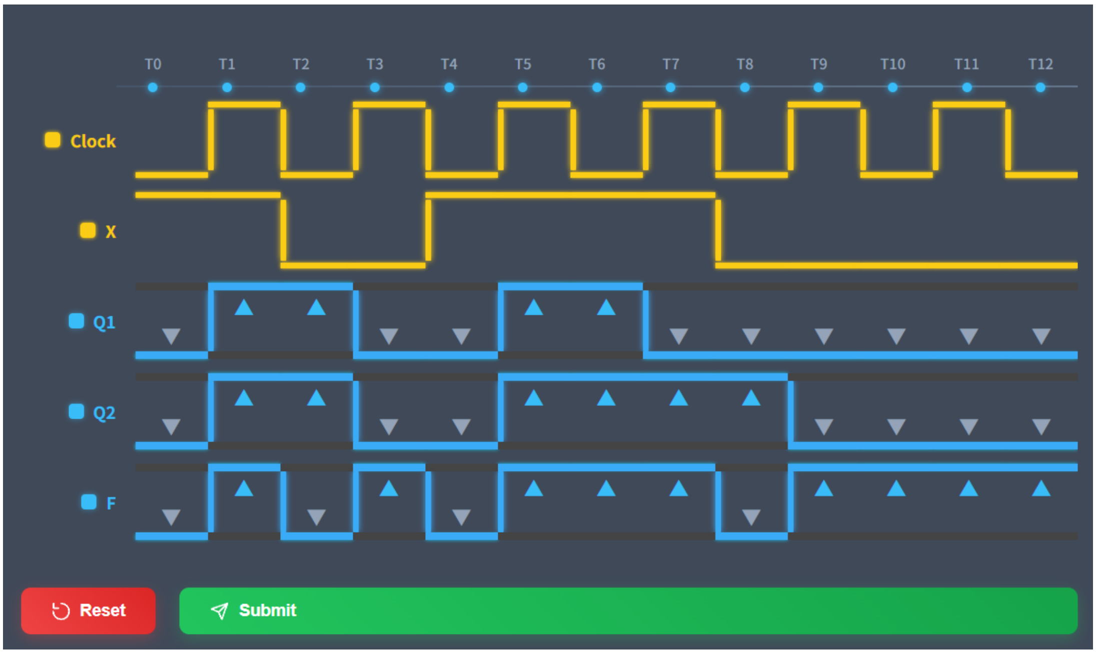


Figure 1: Timing Diagram for Input Sequence X = 101100

4 State Minimization and Analysis (Problem 4)

4.1 1) Minimize the state table

By inspecting the original state table for states with identical outputs and next states (or equivalent next states), we identify the following equivalencies:

- **States** b **and** e**:** Both transition to states (d, c) with outputs $(0, 0)$. Thus, $b \equiv e$. We remove state e.
- **States** d **and** h**:** Both transition to states (g, a) with outputs $(1, 0)$. Thus, $d \equiv h$. We remove state h.
- **States** a **and** c**:** State a transitions to (f, b), and state c transitions to (f, e). Since we established $e \equiv b$, state c is equivalent to a. Thus, $a \equiv c$. We remove state c.

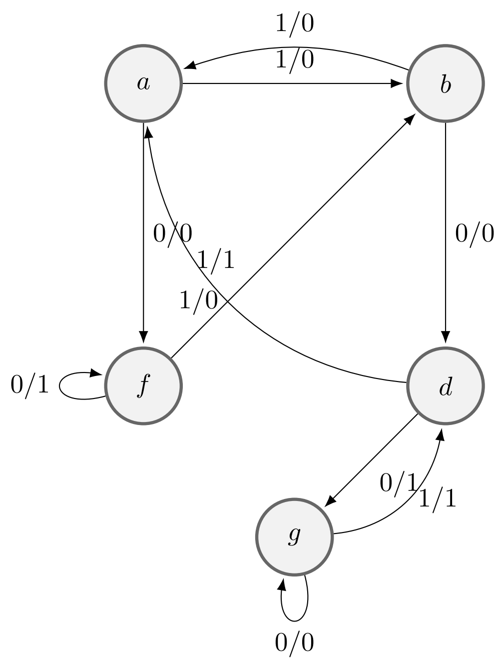
The reduced set of states is $\{a, b, d, f, g\}$.

4.2 2) Reduced State Table

Replacing the removed states (c, e, h) with their equivalents (a, b, d) in the next state columns:

Present State	Next State		Output	
	x=0	x=1	x=0	x=1
a	f	b	0	0
b	d	a	0	0
d	g	a	1	0
f	f	b	1	1
g	g	d	0	1

4.3 3) Reduced State Diagram



4.4 4) Determine the Output Sequence

Input Sequence: 0 1 0 1 0 0 1 0 1 1 1
Starting State: a

We trace the path through the reduced state table:

Time	t_1	t_2	t_3	t_4	t_5	t_6	t_7	t_8	t_9	t_{10}	t_{11}
Input (x)	**0**	**1**	**0**	**1**	**0**	**0**	**1**	**0**	**1**	**1**	**1**
Present State	a	f	b	d	a	f	f	b	d	a	b
Next State	f	b	d	a	f	f	b	d	a	b	a
Output (z)	**0**	**1**	**0**	**0**	**0**	**1**	**1**	**0**	**0**	**0**	**0**

Final Output Sequence: 01000110000

5 Sequential Circuit Design with T Flip-Flops (Problem 5)

5.1 c) Design Procedure

Step 1: Analyze the State Diagram
Identify the transitions for each state (Q_1, Q_2) based on the input x.

Step 2: Construct the State and Excitation Table
For T flip-flops, the excitation input is $T = Q(t) \oplus Q(t+1)$.

- T_{Q1} is the input for flip-flop Q_1.
- T_{Q2} is the input for flip-flop Q_2.

Present State / Input			Next State		Excitation	
Q1	Q2	x	Q1*	Q2*	T_Q1	T_Q2
0	0	0	0	1	0	1
0	0	1	0	0	0	0
0	1	0	1	1	1	0
0	1	1	1	0	1	1
1	0	0	1	1	0	1
1	0	1	1	0	0	0
1	1	0	0	0	1	1
1	1	1	1	1	0	0

Step 3: K-Map Minimization
1. K-Map for T_{Q1}:
We map the minterms where $T_{Q1} = 1$: m_2, m_3, m_6.

Q_2x	00	01	11	10
$Q_1 = 0$	0	0	1	1
$Q_1 = 1$	0	0	0	1

Groups:

- Red Group (m_2, m_3): $(Q_1)'Q_2$
- Blue Group (m_2, m_6): $Q_2(x)'$

Equation for T_{Q1}:

$$T_{Q1} = (Q_1)'Q_2 + Q_2(x)' = \mathbf{Q_2((Q_1)' + (x)')}$$

2. K-Map for T_{Q2}:
We map the minterms where $T_{Q2} = 1$: m_0, m_3, m_4, m_6.

	00	01 Q_2x 11		10
$Q_1 = 0$	1	0	1	0
$Q_1 = 1$	1	0	0	1

Groups:

- Red Group (m_0, m_4): $(Q_2)'(x)'$
- Blue Group (m_4, m_6 - wrap around): $Q_1(x)'$
- Green Group (m_3): $(Q_1)'Q_2x$

Equation for T_{Q2}:

$$T_{Q2} = \underbrace{(Q_2)'(x)' + Q_1(x)'}_{\text{Factor } (x)'} + (Q_1)'Q_2x$$

$$\mathbf{T_{Q2} = (x)'(Q_1 + (Q_2)') + x((Q_1)'Q_2)}$$

Note: The equation for T_{Q2} can be further implemented as an XNOR gate logic: $T_{Q2} = x \odot ((Q_1)'Q_2)$, but the SOP form above is sufficient.

6 Design of Specific Flip-Flop (Problem 6)

6.1 Design Procedure

Step 1: Characteristic Table
Based on the function table provided in the problem (Load when $G = 1$, Hold when $G = 0$), we construct the characteristic table. Since we are using a D Flip-Flop, the required excitation input is $D = Q(t + 1)$.

G	**In**	**Q(t)**	**Q(t+1) = D**
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

Step 2: K-Map Simplification
We map the output D into a 3-variable K-Map (G on rows, In, Q on columns).

	00	01 $In\,Q$ 11		10
$G = 0$	0	1	1	0
$G = 1$	0	0	1	1

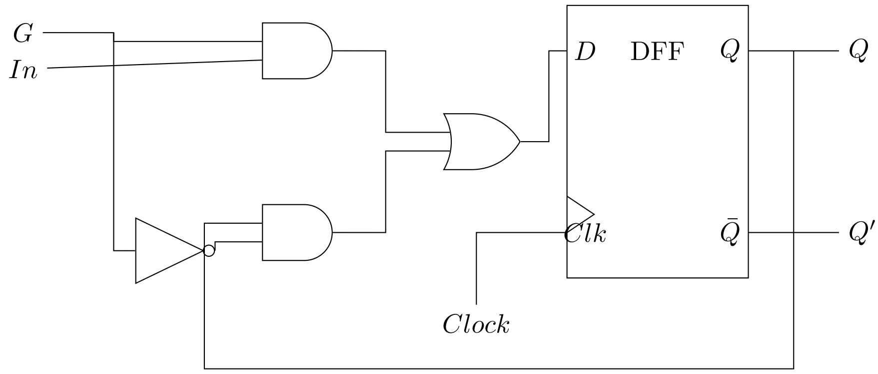
Step 3: Excitation Equation
From the groupings in the K-Map:

- Red Group: $(G)'Q$
- Blue Group: $G \cdot In$

The optimized equation for the D input is:

$$\mathbf{D = G \cdot In + (G)' \cdot Q}$$

6.2 Block Diagram(One possible design)



Problem 7: State Diagram

Machine Type: Mealy Machine.

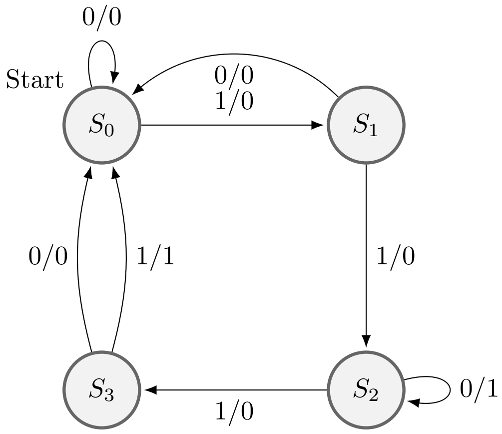


Figure 2: State Diagram.